

PROJECT REPORT

Implementation of DBFuzz for Stored XSS Detection in Doctor Appointment Management System (DAMS)

Submitted By:

Sandeep Khachariya

ID: 2023UCP1628

Dhiraj Chouhan

ID: 2023UCP1591

Submitted To:

(Dr. Ramesh Babu battula)

Department:

(computer science and engineering)

College:

(Malaviya national institute of technology jaipur)

Academic Year:

2025-26

Acknowledgement

We would like to thank our faculty mentor and department for guiding us throughout this project.

We also acknowledge the original DBFuzz research authors for providing the open-source implementation.

Abstract

Web applications are highly vulnerable to security attacks such as Cross-Site Scripting (XSS). Traditional scanners often fail to detect **Stored XSS vulnerabilities**, especially in authenticated areas.

This project implements **DBFuzz**, a database-aware grey-box scanner that detects stored XSS vulnerabilities.

We integrated DBFuzz with a **Doctor Appointment Management System (DAMS)** web application.

The project involved:

- Deploying web application
- Configuring database
- Automating login
- Crawling authenticated pages
- Performing fuzz testing
- Generating logs and results

Additionally, several modifications were made to modernize the outdated research prototype and make it work on Ubuntu.

Table of Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Literature Review
5. System Requirements
6. Architecture
7. Implementation
8. Challenges Faced
9. Novel Work
10. Result
11. Conclusion
12. Future Scope
13. References
- 14.

Chapter 1: Introduction

Web applications store large amounts of user data.

Attackers exploit vulnerabilities such as:

- SQL Injection
- XSS
- CSRF
- Session Hijacking

Among these, **Stored XSS** is dangerous because malicious scripts are stored in the database and executed later.

DBFuzz helps automate detection of such vulnerabilities.

Chapter 2: Problem Statement

Traditional scanners fail because:

- They cannot access authenticated pages
- They do not understand database storage
- They miss hidden endpoints
- They fail to detect stored payload execution

Our goal was to solve this using DBFuzz.

Chapter 3: Objectives

- Implement DBFuzz
 - Deploy DAMS application
 - Configure Docker environment
 - Automate login
 - Perform fuzz testing
 - Analyze results
-

Chapter 4: Literature Review

Research Paper:

Spider-Scents: Grey-box Database-aware Web Scanning for Stored XSS

Presented at USENIX Security Symposium 2024

Repository used:

[DBFuzz GitHub Repository](#)

Chapter 5: System Requirements

Hardware

- 8GB RAM
- Intel i5 processor
- 40GB storage

Software

- Ubuntu Linux
- Python 3.10
- Docker
- Docker Compose
- MariaDB
- phpMyAdmin
- Chromium
- ChromeDriver
- Graphviz

Chapter 6: System Architecture

Architecture Flow

User → DAMS Web Application → Database



DBFuzz



Selenium Login



Cookie Extraction



Crawler



Fuzzer



Logs & Results

Chapter 7: Implementation Steps

Step 1: Clone Repository

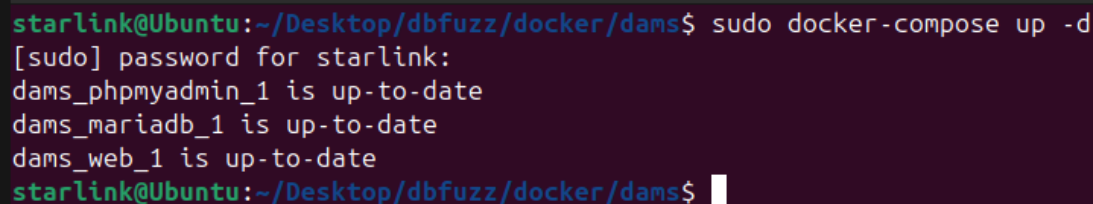
```
git clone https://github.com/Spider-Scents/dbfuzz.git
```

Step 2: Install Dependencies

```
pipenv install
```

Step 3: Deploy DAMS using Docker

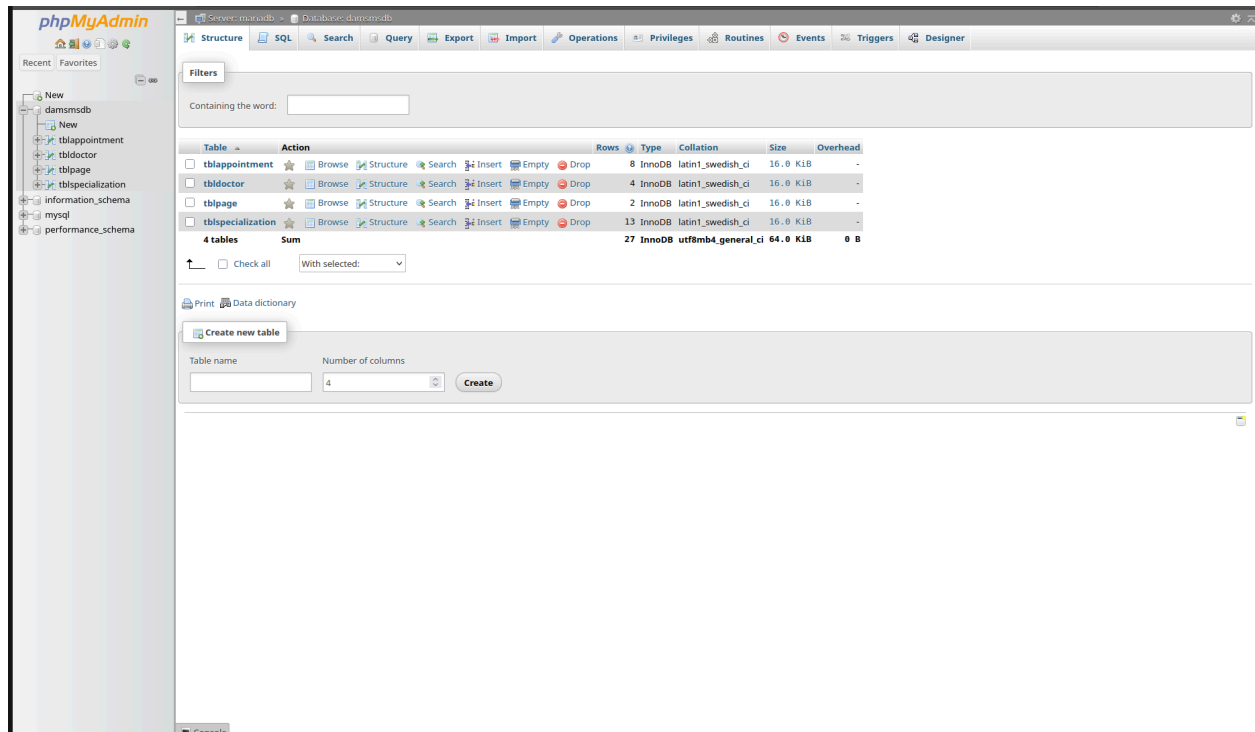
```
cd ~/Desktop/dbfuzz/docker/dams  
sudo docker-compose up -d
```

A terminal window with a dark purple background. The prompt is 'starlink@Ubuntu:~/Desktop/dbfuzz/docker/dams\$'. The command 'sudo docker-compose up -d' is entered. The output shows '[sudo] password for starlink:' followed by three lines: 'dams_phpmyadmin_1 is up-to-date', 'dams_mariadb_1 is up-to-date', and 'dams_web_1 is up-to-date'. The prompt returns to 'starlink@Ubuntu:~/Desktop/dbfuzz/docker/dams\$' with a cursor.

```
starlink@Ubuntu:~/Desktop/dbfuzz/docker/dams$ sudo docker-compose up -d  
[sudo] password for starlink:  
dams_phpmyadmin_1 is up-to-date  
dams_mariadb_1 is up-to-date  
dams_web_1 is up-to-date  
starlink@Ubuntu:~/Desktop/dbfuzz/docker/dams$
```

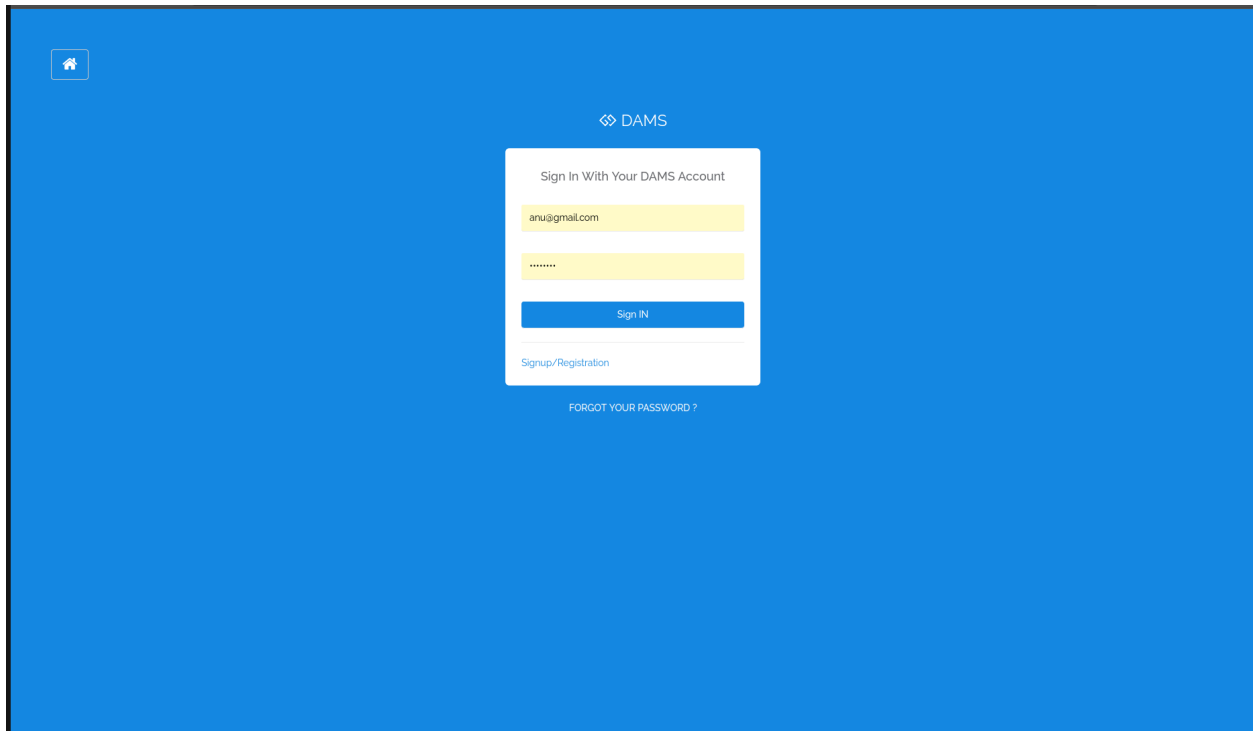
Step 4: Import Database

- Open phpMyAdmin
- Create database **dammsdb**
- Import SQL file



Step 5: Open DAMS Application

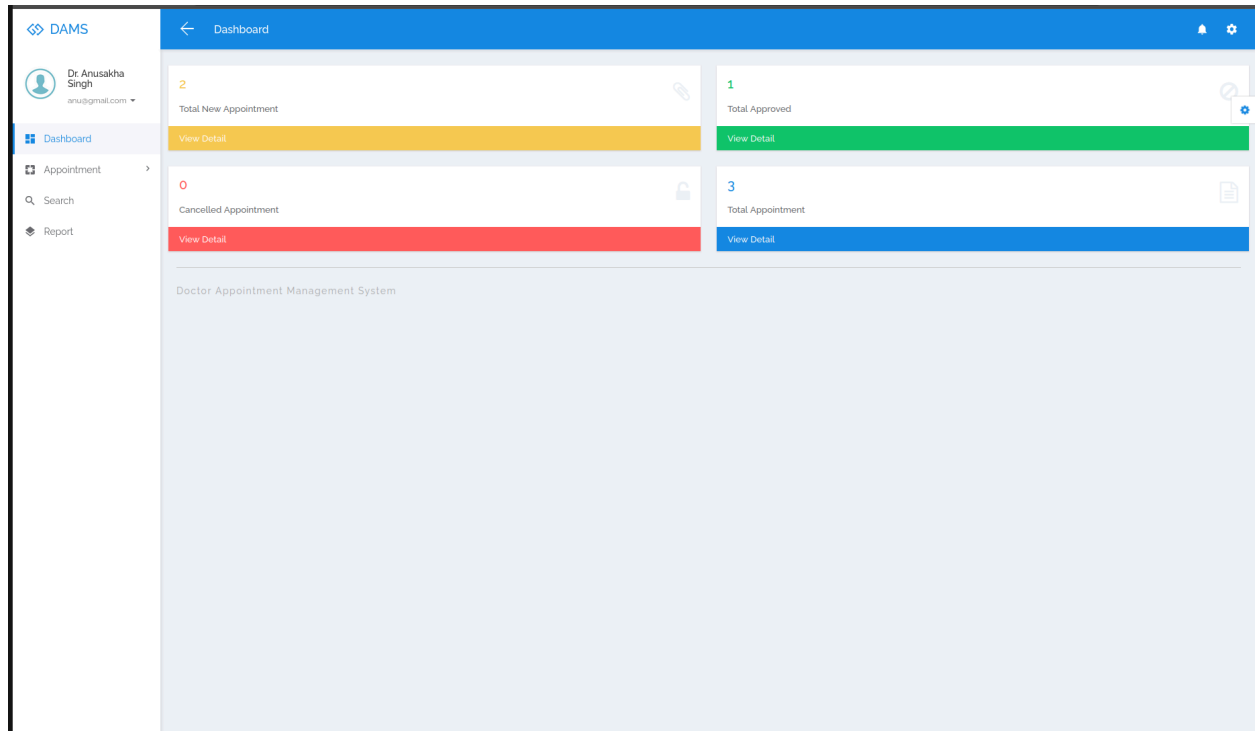
<http://127.0.0.1/dams>



Step 6: Login Credentials

Email: **anu@gmail.com**

Password: **Test@123**



Step 7: Run DBFuzz

```
pipenv run python dbfuzz.py --config docker/dams/config_doctor.ini
```

```

assert len(files) > 0
AssertionError
starlink@Ubuntu: ~/Desktop/dbfuzz$ nano ~/Desktop/dbfuzz/Browser.py
starlink@Ubuntu: ~/Desktop/dbfuzz$ nano ~/Desktop/dbfuzz/Browser.py
starlink@Ubuntu: ~/Desktop/dbfuzz$ cd ~/Desktop/dbfuzz
pipenv run python dbfuzz.py --config docker/dams/config_doctor.ini
fuzzing started at 2026-05-09 15:17:51.463282
BACKWARDS COMPATIBILITY: adding match=0, to credentials
login - url http://localhost/dams/doctor/login.php - credentials match=0,email=anu@gmail.com,password=Test@123
{'download.default_directory': '/home/starlink/Desktop/dbfuzz/cookies'}
Waiting 20 seconds for cookie download
login URL: http://localhost/dams/doctor/cancelled-appointment.php
cookies, user agent, and login URL: {'cookie_str': 'PHPSESSID=095af7707e3384e38ecb28f344c08daf', 'user_agent': 'Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0 Safari/537.36', 'url': 'http://localhost/dams/doctor/cancelled-appointment.php'}
Scanning for list of URLs to check
Initial Scanning http://localhost/dams/doctor/dashboard.php
python3 -u /home/starlink/Desktop/dbfuzz/blackwidow/crawl.py --shallow http://localhost/dams/doctor/dashboard.php --cookies PHPSESSID=095af7707e3384e38ecb28f344c08daf
python3 -u /home/starlink/Desktop/dbfuzz/blackwidow/crawl.py --shallow http://localhost/dams/doctor/dashboard.php --cookies PHPSESSID=095af7707e3384e38ecb28f344c08daf
BW: b'adding cookie  PHPSESSID = 095af7707e3384e38ecb28f344c08daf\n'
BW: b'-----\n'
BW: b'Edges left: 1\n'
BW: b'\n'
BW: b'-----\n'
BW: b'GETS      | FROMS | EVENTS \n'
BW: b'1         | 0     | 0\n'
BW: b'-----\n'
BW: b'Crawl (edge): [get] ROOTREQ -(get None[False])-> [get] http://localhost/dams/doctor/dashboard.php\n'
BW: b'-----\n'
BW: b'Edges left: 24\n'
BW: b'\n'
BW: b'-----\n'
BW: b'GETS      | FROMS | EVENTS \n'
BW: b'14        | 0     | 9\n'
BW: b'-----\n'
BW: b'Crawl (edge): [get] http://localhost/dams/doctor/dashboard.php -(get None[False])-> [get] http://localhost/dams/doctor/search.php\n'
BW: b'-----\n'
BW: b'Edges left: 48\n'
BW: b'\n'
BW: b'-----\n'
BW: b'GETS      | FROMS | EVENTS \n'
BW: b'27        | 0     | 19\n'
BW: b'-----\n'
BW: b'Crawl (edge): [get] http://localhost/dams/doctor/dashboard.php -(get None[False])-> [get] http://localhost/dams/doctor/profile.php\n'
BW: b'-----\n'
BW: b'Edges left: 71\n'
BW: b'\n'
BW: b'-----\n'
BW: b'GETS      | FROMS | EVENTS \n'
BW: b'39        | 0     | 29\n'

```

Chapter 8: Challenges Faced

We encountered several real-world issues:

Docker Variable Issue

Fixed environment variables in docker-compose.

Outdated Debian Repositories

Updated old package sources.

Broken Dockerfile

Rebuilt Dockerfile manually.

Disk Full Issue

Freed storage space.

Chrome Issues

Installed Chromium and ChromeDriver.

Cookie Extension Failure

Original extension was unsupported.

Browser.py Modification

Implemented direct cookie extraction.

Chapter 9: Novel Work

This section is important for marks.

Original project was outdated.

We added:

Docker fixes

Browser fixes

Chromium compatibility fixes

We modernized the original research implementation.

DBFuzz successfully:

- Logged into application
- Crawled authenticated pages
- Discovered endpoints
- Generated logs

Important log file:

blackwidow/logs/crawl-1778320121.7225757.log

```

54', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEB:scantium.webdriver.remote_connection.Finished Request
[{"time": "2026-05-09:15:19:07", "url": "http://blackdown/logs/crawl-1778320121.7225757_log", "method": "GET", "status": 200, "headers": {"Content-Type": "text/plain"}, "body": ""}, {"time": "2026-05-09:15:19:02", "url": "http://localhost/dams/doctor/dashboard.php", "method": "GET", "status": 200, "headers": {"Content-Type": "text/html"}, "body": "

Init crawl on http://localhost/dams/doctor/dashboard.php

 [Classes.py:569] | Init crawl on http://localhost/dams/doctor/dashboard.php || 2026-05-09:15:19:02 | root | INFO | [Functions.py:175] | Check edge: [get] ROOTREQ - (get None[False])-> [get] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:02 | root | INFO | [Functions.py:221] | Follow edge: [get] ROOTREQ - (get None[False])-> [get] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:09 | root | INFO | [Classes.py:1655] | Current url: http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:09 | root | INFO | [Classes.py:1656] | Crawl (edge): [get] ROOTREQ - (get None[False])-> [get] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:13 | root | DEBUG | [Urls.py:120] | URLs from extract urls [get] http://localhost/dams/doctor/appointment-bdwates.php, [get] http://localhost/dams/doctor/dashboard.php, [get] http://localhost/dams/doctor/change-password.php, [javascript] javascript:void(0), [get] http://localhost/dams/doctor/view-appointment-detail.php?editid=4&aaptid=611388102, [get] http://localhost/dams/doctor/dashboard.php#menubar-customizer, [get] http://localhost/dams/doctor/dashboard.php#navbar-customizer, [get] http://localhost/dams/doctor/approved-ap |
| pment.php, [get] http://localhost/dams/doctor/logout.php, [get] http://localhost/dams/doctor/search.php, [get] http://localhost/dams/doctor/view-appointment-detail.php?editid=5&aap |  |  |  |  |
| tid=607441873, [get] http://localhost/dams/doctor/cancelled-appointment.php, [get] http://localhost/dams/doctor/new-appointment.php, [get] http://localhost/dams/doctor/profile.php, [ge |  |  |  |  |
| t] http://localhost/dams/doctor/dashboard.php#, [get] http://localhost/dams/doctor/all-appointment.php) |  |  |  |  |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/appointment-bdwates.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1811] | failed add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/change-password.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/view-appointment-detail.php?editid=4&aaptid=611388102 |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/dashboard.php#menubar-customizer |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/dashboard.php#navbar-customizer |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/approved-appointment.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/logout.php |
| 2026-05-09:15:19:14 | root | DEBUG | [Functions.py:337] | Different origins [http://localhost/dams/doctor/dashboard.php and http://localhost/dams/doctor/logout.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1812] | Not allowed to add edge: [get] http://localhost/dams/doctor/dashboard.php -(get None[False])-> [get] http://localhost/dams/doctor/logout.php |
| ns/doctor/logout.php |  |  |  |  |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/search.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/view-appointment-detail.php?editid=5&aaptid=607441873 |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/cancelled-appointment.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/new-appointment.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/profile.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1806] | from URLs [get] http://localhost/dams/doctor/all-appointment.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:1828] | from events [event] http://localhost/dams/doctor/dashboard.php |
| 2026-05-09:15:19:14 | root | INFO | [Classes.py:135] | failed to add node [event] http://localhost/dams/doctor/dashboard.php, already added |
| 2026-05-09:15:19: |

```

Error Logs

cat errors.log

```
starlink@Ubuntu: ~/Desktop/dbfuzz$ cat errors.old.log
DEBUG:selenium.webdriver.common.service:Started executable: 'chromedriver' in a child process with pid: 51103
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session {"capabilities":{"firstMatch":[{"browserName":"chrome","pageLoadStrategy":"normal","goog:chromeOptions":{"prefs":{"download.default_directory":"/home/starlink/Desktop/dbfuzz/cookies"},"extensions":[],"binary":"/usr/bin/chromium-browser","args":["-no-sandbox","-disable-dev-shm-usage","-remote-debugging-port=9222","load-extension=/home/starlink/Desktop/dbfuzz/cookie_extension"]}]},"alwaysMatch":{"browserName":"chrome","pageLoadStrategy":"normal","goog:chromeOptions":{"prefs":{"download.default_directory":"/home/starlink/Desktop/dbfuzz/cookies"},"extensions":[],"binary":"/usr/bin/chromium-browser","args":["-no-sandbox","-disable-dev-shm-usage","-remote-debugging-port=9222","load-extension=/home/starlink/Desktop/dbfuzz/cookie_extension"]}]}
DEBUG:urllib3.connectionpool:Starting new HTTP connection (1): localhost:43789
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session HTTP/1.1" 200 883
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":{"capabilities":{"acceptInsecureCerts":false,"browserName":"chrome","browserVersion":"147.0.7727.116","chrome":{"chromedriverVersion":"147.0.7727.116 (dbcf1b1bfb506cc580859bcb5ff9460a8443af90-refs/branch-heads/7727_1110/#3)}","userDataDir":"/tmp/org.chromium.Chromium.scoped_dir.YBYQT"},"fedcm:accounts":true,"goog:chromeOptions":{"debuggerAddress":"localhost:9222"},"goog:processID":51165,"networkConnectionEnabled":false,"pageLoadStrategy":"normal","platformName":"linux","proxy":{},"setWindowRect":true,"strictFileInteractability":false,"timeouts":{"implicit":0,"pageLoad":300000,"script":30000},"unhandledPromptBehavior":"dismiss and notify","webauthn:extension:credBlob":true,"webauthn:extension:largeBlob":true,"webauthn:extension:minPinLength":true,"webauthn:extension:prf":true,"webauthn:virtualAuthenticators":true},"sessionId":"7dd04d91dbdd999a57243fca81dd524f"}} | headers=HTTPHeaderDict({'Content-Length': '883', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/window/rect {"x": 200, "y": 10, "width": null, "height": null}
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session/7dd04d91dbdd999a57243fca81dd524f/window/rect HTTP/1.1" 200 52
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":{"height":1180,"width":945,"x":200,"y":10}} | headers=HTTPHeaderDict({'Content-Length': '52', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/url {"url": "http://localhost/dams/doctor/login.php"}
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session/7dd04d91dbdd999a57243fca81dd524f/url HTTP/1.1" 200 14
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":null} | headers=HTTPHeaderDict({'Content-Length': '14', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/execute/sync {"script": "return navigator.userAgent", "args": []}
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session/7dd04d91dbdd999a57243fca81dd524f/execute/sync HTTP/1.1" 200 113
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"} | headers=HTTPHeaderDict({'Content-Length': '113', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/elements {"using": "css selector", "value": "[name='email']"}
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session/7dd04d91dbdd999a57243fca81dd524f/elements HTTP/1.1" 200 127
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":[{"element-6066-11e4-a52e-4f735466cecf":"f.CB8E7902B361B4B97A492DE54EC58576.d.E8E10DF28F1796396545C29AF21A6C15.e.2"}]} | headers=HTTPHeaderDict({'Content-Length': '127', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/element/f.CB8E7902B361B4B97A492DE54EC58576.d.E8E10DF28F1796396545C29AF21A6C15.e.2/value {"text": "anu@gmail.com", "value": ["a", "n", "u", "a", "t", "a", "i", "l", ".", "c", "o", "m"], "id": "f.CB8E7902B361B4B97A492DE54EC58576.d.E8E10DF28F1796396545C29AF21A6C15.e.2"}
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session/7dd04d91dbdd999a57243fca81dd524f/element/f.CB8E7902B361B4B97A492DE54EC58576.d.E8E10DF28F1796396545C29AF21A6C15.e.2/value HTTP/1.1" 200 14
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":null} | headers=HTTPHeaderDict({'Content-Length': '14', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/elements {"using": "css selector", "value": "[name='password']"}
DEBUG:urllib3.connectionpool:http://localhost:43789 "POST /session/7dd04d91dbdd999a57243fca81dd524f/elements HTTP/1.1" 200 127
DEBUG:selenium.webdriver.remote.remote_connection:Remote response: status=200 | data={"value":[{"element-6066-11e4-a52e-4f735466cecf":"f.CB8E7902B361B4B97A492DE54EC58576.d.E8E10DF28F1796396545C29AF21A6C15.e.3"}]} | headers=HTTPHeaderDict({'Content-Length': '127', 'Content-Type': 'application/json; charset=utf-8', 'cache-control': 'no-cache'})
DEBUG:selenium.webdriver.remote.remote_connection:Finished Request
DEBUG:selenium.webdriver.remote.remote_connection:POST http://localhost:43789/session/7dd04d91dbdd999a57243fca81dd524f/element/f.CB8E7902B361B4B97A492DE54EC58576.d.E8E10DF28F1796396545
```

Discovered Endpoints

- dashboard.php
- search.php
- profile.php
- appointment pages

Chapter 11: Conclusion

This project successfully implemented DBFuzz on DAMS.

We reproduced the original research workflow and improved compatibility with modern systems.

Chapter 12: Future Scope

- Add AI vulnerability analysis
 - Better dashboards
 - Support more applications
 - Improve automation
-

Chapter 13: References

[DBFuzz GitHub Repository](#)

USENIX Security Symposium 2024 Research Paper

Docker documentation

Python documentation